International Journal of Infectious Diseases 15 (2011) e167–e173

Contents lists available at ScienceDirect

International Journal of Infectious Diseases

journal homepage: www.elsevier.com/locate/ijid

Invasive *Haemophilus influenzae* in British Columbia: non-Hib and non-typeable strains causing disease in children and adults

Michelle Shuel [a], Linda Hoang [b,c], Dennis K.S. Law [a], Raymond Tsang [a,*]

[a] *Vaccine Preventable Bacterial Diseases, National Microbiology Laboratory, Public Health Agency of Canada, 1015 Arlington Street, Winnipeg, Manitoba, Canada R3E 3R2*
[b] *British Columbia Center for Disease Control, Public Health Microbiology and Reference Laboratory, Vancouver, British Columbia, Canada*
[c] *Department of Pathology and Laboratory Medicine, University of British Columbia, Vancouver, British Columbia, Canada*

ARTICLE INFO

Article history:
Received 7 June 2010
Received in revised form 12 October 2010
Accepted 13 October 2010

Corresponding Editor: Timothy Barkham, Tan Tock Seng, Singapore

Keywords:
Haemophilus influenzae
Serotype
MLST
Antibiotic resistance

SUMMARY

Objectives: To characterize invasive *Haemophilus influenzae* and to examine the population at risk for invasive *H. influenzae* disease in British Columbia, Canada, 2008–2009.
Methods: H. influenzae recovered from individual patients were characterized by serotyping, biotyping, multilocus sequence typing (MLST), and antibiotic susceptibility testing. Age information was recorded from specimen requisition forms.
Results: Of the 98 cases, 66% were caused by non-typeable strains, followed by serotypes b (12%), a (10%), f (10%), and e (1%). Cases caused by serotypes b and f and non-typeable strains were mainly in adults over 18 years of age, while cases due to serotype a were mainly in children under the age of 2 years. Different sequence types were found in encapsulated strains according to their serotypes, and non-typeable strains had their own unique sequence types. No capsule switching was documented. Antibiotic resistance was common among non-typeable strains, with 31% identified as genotypic β-lactamase-negative ampicillin-resistant (BLNAR) strains.
Conclusion: Invasive *H. influenzae* disease in a population vaccinated against Hib was age-dependent and involved both non-typeable and encapsulated strains. Adults were susceptible to invasive diseases due to non-typeable and serotype b and f strains, while in children, most diseases were due to serotype a bacteria.

Crown Copyright © 2010 Published by Elsevier Ltd on behalf of International Society for Infectious Diseases. All rights reserved.

1. Introduction

Haemophilus influenzae is a strict human respiratory tract pathogen that can cause a spectrum of mild (otitis media, sinusitis) to severe (meningitis, epiglottitis, cellulitis, septicemia) disease. Six different serotypes have been identified based on the serological specificities of their polysaccharide capsules,[1] with capsular serotype b as the predominant strain associated with invasive *H. influenzae* disease. Strains without capsules are referred to as non-typeable.

Before the introduction of vaccines against *H. influenzae*, invasive *H. influenzae* disease was mainly a childhood disease, with most cases found in children under the age of 10 years and caused mostly by serotype b organisms.[2,3] Vaccination against *H. influenzae* serotype b (Hib) was introduced in the 1980s, first with the plain polysaccharide vaccine, and soon followed by conjugated vaccines in the early 1990s, which induce T-cell-mediated long-term immunity and are immunogenic when given to infants.[4] Dramatic reductions in invasive *H. influenzae* disease were reported shortly after countries introduced Hib conjugate vaccines into their routine childhood immunization programs. In the USA, elimination of Hib disease was considered a possible goal.[5] In Canada, a historic low in the number of invasive Hib disease was recorded as early as 2000 (with only four cases found by a network of 12 major pediatric centers across the country), eight to nine years after the Hib conjugate vaccine was introduced.[6]

Before Hib conjugate vaccines were widely used, non-Hib *H. influenzae* (either non-encapsulated strains or other serotypes) were not regarded as significant causes of invasive disease.[2,3,7] In theory, vaccines that target only one serotype may drive the organism to either switch and/or replace their capsules with non-vaccine capsular types or shed their capsule altogether to become resistant to the vaccine.[8–10] Capsule switching involves strains with capsule types included in the vaccine acquiring capsule synthesis genes from strains with capsule types not included in the vaccine, to become resistant to the vaccine while still retaining their original virulence characters. Capsule replacement occurs when strains with capsule types not included in the vaccine emerge and replace strains with capsule types included in the vaccine to cause disease.[9] Indeed, non-Hib *H. influenzae*, which

* Corresponding author. Tel.: +1 204 789 6020; fax: +1 204 789 2130.
E-mail address: Raymond.tsang@phac-aspc.gc.ca (R. Tsang).

1201-9712/$36.00 – see front matter. Crown Copyright © 2010 Published by Elsevier Ltd on behalf of International Society for Infectious Diseases. All rights reserved.
doi:10.1016/j.ijid.2010.10.005

used to be insignificant, has now become a larger player in causing invasive disease.[11–14]

In the past, surveillance of invasive *H. influenzae* disease had been restricted to monitoring only Hib cases. In the last several years, the National Microbiology Laboratory (NML) has begun to monitor invasive *H. influenzae* disease by characterizing strains recovered from invasive disease cases, regardless of their capsule status or capsular type. However, this work has been restricted to only one region of the country, namely Manitoba and Northwestern Ontario.[12,13] A recent study in the province of Ontario has confirmed that strain replacement is happening in invasive *H. influenzae* disease.[14] Canada has a relatively small population of about 34 million in relation to its territorial size of about 9.9 million square kilometers or 3.8 million square miles. Previous studies on *H. influenzae* have mainly focused on isolates recovered from two neighboring provinces of Manitoba and Ontario, situated in the centre to eastern part of the country, respectively. In order to gain a wider perspective of this invasive bacterial disease and to study the evolving nature of the causative agent, we extended our study to British Columbia, which is located in the most western part of the country, about 3000 km or 2000 miles from Ontario. In this study, the British Columbia Center for Disease Control (BCCDC) Public Health Microbiology and Reference Laboratory in collaboration with the NML examined invasive disease cases and characterized the *H. influenzae* isolates obtained in the last 2 years. The objectives of this study were to examine (1) the biology of the current invasive *H. influenzae* strains, and (2) the population currently at risk for invasive disease, after more than 15 years of Hib conjugate vaccine usage in this country.

2. Materials and methods

2.1. Bacterial isolates

From January 1, 2008 to December 31, 2009, there were 98 strains of invasive *H. influenzae* isolates submitted to the British Columbia Center for Disease Control's Public Health Microbiology and Reference Laboratory for serotyping as part of the routine practice. All isolates were from individual cases with invasive disease as determined by the submitting laboratory, and duplicate isolates from the same patient were excluded from this study. One patient had two separate episodes of infection due to a nontypeable strain that occurred several months apart, and isolates from each infection have been included. Identities of all isolates were confirmed by standard biochemical tests[15] and 16S rRNA sequencing.[16] Biotypes were assigned according to Kilian's biotyping scheme based on three biochemical reactions: urease, indole production, and ornithine decarboxylase.[17]

2.2. Serotyping

The serotypes were determined by slide agglutination test using antisera against all six serotypes purchased from commercial sources (Difco, Oakville, Ontario, Canada; Denka Seiken, Tokyo, Japan). Serotyping by PCR detection of the serotype-specific genes and the capsule transport, *bexA*, gene was carried out using primers described by Falla et al.[18]

2.3. Multilocus sequence typing

Multilocus sequence typing (MLST) was carried out by PCR amplification of seven housekeeping genes (*adk*, *atpG*, *frdB*, *fucK*, *mdh*, *pgi* and *recA*) according to the method previously described,[19] and the assignment of sequence types (STs) was done using the *H. influenzae* MLST website (http://haemophilus.mlst.net/).

2.4. Antibiotic resistance and β-lactamase production

β-Lactamase production was detected using DrySlide Nitrocefin (BBL, Becton Dickinson, Oakville, Ontario, Canada). Disk diffusion testing was carried out as described by the Clinical and Laboratory Standards Institute (CLSI).[20] The following antibiotics were tested: ampicillin, amoxicillin–clavulanic acid, cefaclor, ceftriaxone, chloramphenicol, ciprofloxacin, clarithromycin, levofloxacin, moxifloxacin, trimethoprim–sulfamethoxazole, and tetracycline (Oxoid, Nepean, Ontario, Canada). In addition, a 2-μg ampicillin disk was also used to detect β-lactamase-negative isolates showing reduced susceptibility to ampicillin.[21] β-Lactamase-negative ampicillin-resistant (BLNAR) *H. influenzae* strain ATCC 49247 was used as a control in each experiment.

2.5. Determination of BLNAR genotype by ftsI gene sequencing

PCR amplification of the *ftsI* gene, which encodes penicillin-binding protein 3 (PBP3), was carried out on all strains. Previously described primers and conditions were used to amplify approximately 1 kb of the 1.8-kb *ftsI* gene.[22] The resulting PCR products were purified using the QIAquick PCR purification kit and sequenced. Results were compared to a previously published GenBank sequence of a β-lactamase-negative ampicillin-susceptible (BLNAS) *H. influenzae* strain to determine any amino acid substitutions (GenBank accession number **L42023**).[23] Strains with *ftsI* gene mutations that led to amino acid substitutions in their PBP3 were classified into groups according to the nomenclature used by Ubukata et al.[22] and Dabernat et al.[24]

3. Results

3.1. Invasive H. influenzae cases in British Columbia, 2008–2009

There were 53 individual case isolates in 2008 and 45 in 2009. Age information for the 98 cases is presented in Figure 1A. Overall, 79% of the cases ($n = 77$) occurred in individuals over the age of 10 years, and only 21% of the cases ($n = 21$) were in children under 11 years of age; 17% ($n = 17$) were in children aged 2 years or younger.

When cases were broken down into the responsible serotypes, significant differences were found (Figure 1B). For cases due to Hib, 25% (3/12 cases) were found in children aged 2 years or below and 75% (9/12 cases) were found in adults aged 21 years or above. In contrast, 80% (8/10 cases) of the serotype a cases were in children under 2 years, and there were only two adult cases, in the age group ranges of 51–60 years and 61–70 years. Similar to Hib, most of the cases due to non-typeable *H. influenzae* were in adults over the age of 20 years (54/65 cases; 83%); 9% (6/65 cases) of the non-typeable cases were in children aged under 2 years and another 5% of the cases were in children aged 3–10 years. Unique to the serotype f cases was that with the exception of one child (7 years old), only adults were affected.

Most of the 98 invasive isolates ($n = 80$; 82%) were recovered from blood cultures. There were only 10 cases with *H. influenzae* recovered from the cerebrospinal fluid, and in three of these 10 cases, the isolates were also found in blood cultures. The remaining invasive case isolates were recovered from various body sites, as shown in Table 1.

3.2. Serotypes and biotypes of invasive H. influenzae in British Columbia

Non-typeable strains accounted for most of the cases ($n = 65$; 66%), followed by serotype b ($n = 12$; 12%) and serotypes a ($n = 10$; 10%) and f ($n = 10$; 10%); the remaining case was due to a serotype e strain (Table 1).

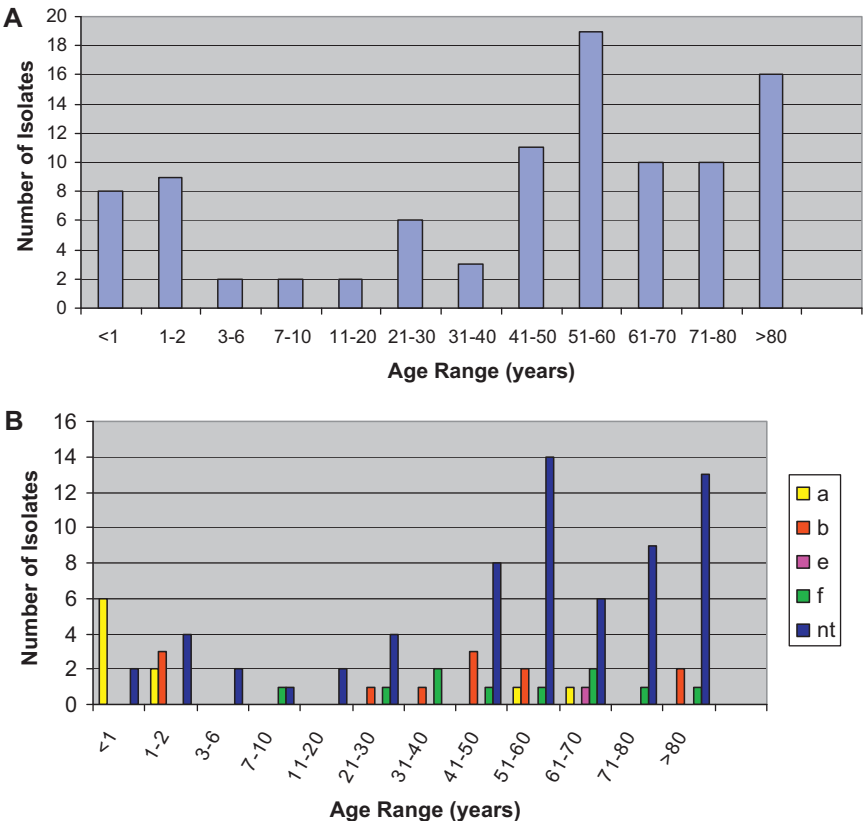


Figure 1. (A) Distribution of the 98 invasive *Haemophilus influenzae* isolates from British Columbia by patient age range. (B) Serotype distribution by age range for the 98 invasive *Haemophilus influenzae* isolates from British Columbia.

All serotype a isolates belonged to biotype II. The single serotype e isolate, 10 of the serotype b isolates, and nine serotype f isolates were all identified as biotype I. Two serotype b and one serotype f were found to be biotype II. Biotypes I, II, III, IV, V, VI, and VII were found among the non-typeable isolates, with biotype II being the most common biotype found (58%).

3.3. Clonal analysis of invasive H. influenzae from British Columbia

None of the 33 serotypeable isolates were found to have a ST identical or related to the STs found among the non-typeable isolates (data not shown). Each serotype was made up of isolates with unique STs and the STs observed were not found in more than one serotype (Table 2). Of the 12 STs identified in the 33 serotypeable isolates, only two identical alleles were found among isolates belonging to different serotypes. One serotype b isolate (ST-53) shared an identical *frdB* allele with serotype a isolates belonging to three different STs. Similarly, a single serotype f isolate (ST-16) was found to have an *atpG* allele identical to that in the single serotype e isolate (ST-66). The non-typeable isolates were divided into 40 unique STs and showed a high degree of diversity in their seven housekeeping gene allelic profiles (Figure 2).

Table 1
Serotypes and sources of isolation of 98 invasive *Haemophilus influenzae* isolates collected in British Columbia, 2008–2009

Source	Serotype				
	a	b	e	f	Non-typeable
Blood	5	11	1	8	55
CSF	3	1		2	1
Blood and CSF	2				1
Lung					2
Brain					1
Meninges					1
Pelvic abscess					1
Eye socket					1
Pleural fluid					1
Tissue: product of conception					1
Total	10	12	1	10	65

CSF, cerebrospinal fluid.

3.4. Antibiotic susceptibility of invasive H. influenzae in British Columbia

The antibiotic resistance profiles, production of β-lactamase, and *ftsI* gene mutations for the 98 invasive *H. influenzae* isolates are shown in Table 3. There was no ampicillin resistance due to β-lactamase production seen among the 33 serotypeable isolates. The only observed resistance among the serotypeable isolates was two isolates showing resistance to trimethoprim–sulfamethoxazole (one serotype b and one serotype f).

On the other hand, resistance to antibiotics was commonly found among the non-typeable isolates. Thirteen of the 65 (20%) non-typeable isolates were found to produce β-lactamase and were resistant to ampicillin. In addition, 20 (31%) of the non-typeable isolates contained significant mutations in their *ftsI* genes that led to amino acid substitutions occurring around the serine–serine–asparagine (SSN) and lysine–threonine–glycine (KTG) motifs of their penicillin-binding protein.[22] Mutations near the conserved SSN and KTG regions were associated with decreased affinity to ampicillin; these mutations have been reported in known BLNAR strains.[22,24] Based on their *ftsI* gene sequences, 13 of the 20 non-typeable isolates were classified as group IIb, four as group IId, and one each as group I, group IIa, and group III. The amino acid substitutions found among these 20 non-typeable isolates are shown in Table 4.

Table 2
Allele profile of housekeeping genes and sequence types as determined by multilocus sequence typing of serotypeable strains of invasive *Haemophilus influenzae* in British Columbia, 2008–2009

Serotype	No. of isolates	Housekeeping gene allele number							Sequence type
		adk	*atpG*	*frdB*	*fucK*	*mdh*	*pgi*	*recA*	
a	5	13	16	5	18	3	11	36	576
	3	13	16	5	18	3	11	7	56
	2	13	16	5	2	3	11	7	23
b	8	42	14	4	5	4	7	8	231
	1	10	14	4	5	4	7	8	6
	1	10	14	4	5	85	7	8	190
	1	10	14	5	7	4	7	8	53
	1	28	14	4	3	4	3	8	157
e	1	18	6	3	11	10	28	12	66
f	8	22	19	11	11	22	19	15	124
	1	22	4	11	11	22	19	15	598
	1	22	6	11	12	1	18	15	16

Many of these genotypic BLNAR isolates were also found to have resistance to multiple antibiotics (Table 3). Half (10/20) of the genotypic BLNAR strains showed some degree of resistance to ampicillin, e.g., resistance to the 2 μg ampicillin disk and/or intermediate to the 2 and 10 μg ampicillin disk. Three of the 13 non-typeable β-lactamase producing isolates had similar significant *ftsI* mutations and were therefore regarded as genotypic β-lactamase-positive amoxicillin–clavulanic acid-resistant (BLPACR) isolates.

4. Discussion

Although several prior publications have documented changes in the epidemiology of invasive *H. influenzae* disease in the USA, Canada, and Europe,[11,12,25] few have provided details on the biology of the evolving isolates. Among the five different STs identified in the 12 invasive Hib isolates in British Columbia, only one ST (ST-190) was also found among the five Hib isolates analyzed in Manitoba.[26] Of the 17 Hib isolates typed in these two provinces (British Columbia and Manitoba), seven different STs were found with 15 different alleles detected in the seven housekeeping genes analyzed. The most varied housekeeping gene was *adk* (four alleles), followed by *fucK* (three alleles), and *frdB*, *mdh*, and *pgi* (each with two different alleles). No variations were found in the *atpG* or *recA* genes.

In British Columbia, ST-576 was the most common ST identified in serotype a isolates (5/10 isolates), while in Manitoba, most serotype a isolates (35/36 isolates) were typed as ST-23.[26] ST-576 and ST-23 are highly related, with six of their seven housekeeping gene alleles being identical. The degree of heterogeneity among the

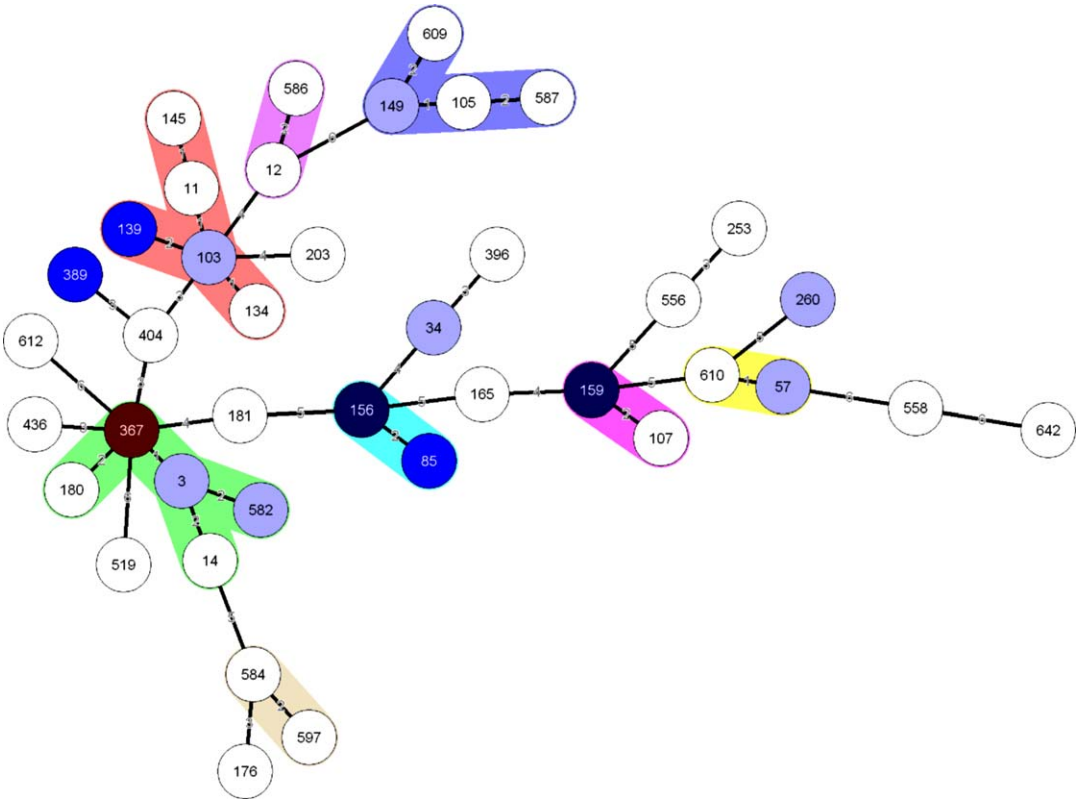


Figure 2. Minimum spanning tree diagram showing the multilocus sequence typing results for 65 invasive non-typeable *Haemophilus influenzae* strains from British Columbia. Sequence types are listed within each node. Sequence distances (the number of non-identical alleles between sequence types (STs)) are listed between each node and the clonal complexes are indicated by the shaded backgrounds. Node color: white = 1 isolate; light blue = 2 isolates; royal blue = 3 isolates; dark blue = 4 isolates; red = 7 isolates.

Table 3
Antibiotic resistance profiles and amino acid substitutions in penicillin-binding protein 3 (PBP3) determined by *ftsI* gene sequencing in invasive *Haemophilus influenzae* from British Columbia, 2008–2009

Serotype	No. of isolates	β-Lactamase production	Antibiotic resistance profile[a]	PBP3 mutations
a	10	–	Susceptible to all	No
b	1	–	Cec I	No
	1	–	Sxt R	No
	10	–	Susceptible to all	No
e	1	–	Susceptible to all	No
f	1	–	Sxt R	No
	9	–	Susceptible to all	No
nt	5	+	Amp R	No
	1	+	Amp R, Cec I	No
	2	+	Amp R, Clr I	No
	1	+	Amp R, Sxt R	No
	1	+	Amp R, Amc R	No
nt	4	–	Sxt R	No
	28	–	Susceptible to all	No
nt BLPACR genotype	1	+	Amp R	Yes
	1	+	Amp R, Clr I	Yes
	1	+	Amp R, Chl R, Sxt R, Tet R	Yes
nt BLNAR genotype	2	–	Amp I (2 μg)	Yes
	4	–	Amp I (2 and 10 μg)	Yes
	1	–	Amp R (2 μg), Amp I (10 μg)	Yes
	1	–	Amp R (2 μg), Amp I (10 μg), Cec I	Yes
	1	–	Amp I (2 μg), Cec R, Sxt R	Yes
	1	–	Amp R, Amc R	Yes
	1	–	Cec R, Clr R	Yes
	5	–	Sxt R	Yes
	4	–	Susceptible to all	Yes

nt, non-typeable; BLPACR, β-lactamase-positive amoxicillin–clavulanic acid-resistant; BLNAR, β-lactamase-negative ampicillin-resistant; R, resistant; I, intermediate; Amc, amoxicillin–clavulanic acid; Amp, ampicillin; Cec, cefaclor; Chl, chloramphenicol; Clr, clarithromycin; Sxt, trimethoprim–sulfamethoxazole; Tet, tetracycline.
[a] Determined by disk diffusion according to Clinical and Laboratory Standards Institute (CLSI) guidelines.

46 serotype a isolates from these two provinces was less than that found among the Hib isolates. Variations in serotype a isolates were only found in two housekeeping genes: *recA* (three alleles) and *fucK* (two alleles), with the rest showing complete identity between the 46 isolates studied. Even less variation was found among the serotype f isolates in these two provinces: ST-124 was the most common ST in British Columbia and was the only ST found in the five serotype f isolates in Manitoba.

Our results from studies of Hib in British Columbia and Manitoba appear to confirm an earlier report of increased genetic diversity of Hib in the post-Hib vaccination era.[27] In contrast, the clonal nature of serotype f isolates recovered in two different provinces in Canada also support the clonal relationship of invasive serotype f isolates recovered from Denmark and the USA.[28]

Serotype a *H. influenzae* infections have been frequently found in aboriginal populations in both the USA[29,30] and Canada.[13,31] However, detailed genetic comparisons of strains recovered from different localities have not been carried out, and its evolution in response to Hib vaccine pressure is important to follow, as some regard serotype a strains to be similar to Hib in terms of their pathogenic potential.[32,33]

With the exception of serotype a isolates belonging to biotype II, most other serotypeable strains were of biotype I, which reflects their limited genetic diversity when studied by MLST. In contrast, seven biotypes were found among the non-typeable strains. Similarly, among the serotypeable strains, where strains of the same serotype had most of their alleles identical, many of the non-typeable strains had most of their alleles different (as many as seven non-identical alleles). There were eight clonal complexes found among the non-typeable *H. influenzae*, ranging from groups of two to 13 isolates. The invasive non-typeable isolates from Manitoba also displayed a high degree of diversity. In this study, we found 40 unique STs among the 65 non-typeable isolates in British Columbia, and in Manitoba, we observed 45 unique STs among 68 isolates.[26] There were only 14 STs common to both provinces. The most common ST in British Columbia was ST-367 (seven isolates), and this ST was not seen in Manitoba; however related STs (ST-180, ST-3, and ST-14) made up 21% of the Manitoba isolates. ST-14 was most common in Manitoba (eight isolates) and is different from ST-367 by three alleles.

The serotype and MLST analysis of invasive *H. influenzae* suggests an evolving population of bacteria, possibly as a result of

Table 4
Amino acid substitutions in penicillin-binding protein 3 of β-lactamase-negative non-typeable *Haemophilus influenzae*

Group[a]	No. of strains	Amino acid substitution														
I	1										R517H					
IIa	1	D350N										N526K				
IIb	2									A502V		N526K				
	5	D350N		M377I						A502V		N506K				
	4	D350N		M377I						A502V		N526K		V547I		
	1	D350N					A437S			A502V		N526K		V547I		
	1	D350N							G490E			N526K	A530S			
IId	1							I449V				N526K				
	3							I449V				N526K		V547I		
III	1	D350N	S357N	M377I	S385T	L389F						N526K		V547I		V562L

[a] According to the classification scheme proposed by Ubukata et al.[22] and Dabernat et al.[24]

adaptation in response to Hib vaccine pressure and immune selection from the host. This study based on 98 invasive *H. influenzae* isolates recovered recently in British Columbia confirms our earlier observation of no detected capsule switching;[26] however, non-typeable *H. influenzae* has become the predominant causative agent of invasive *H. influenzae* disease. Although the highly heterogeneous nature of non-typeable *H. influenzae* with many clonal types causing invasive disease may not suggest that disease is due to evolution of virulent clones, the lack of clinical information on the patients prevents a more definite statement on this issue, especially when four clones were responsible for almost half of all the invasive cases due to non-typeable strains in British Columbia.

While none of the 33 invasive encapsulated *H. influenzae* in British Columbia were found to be ampicillin-resistant due to β-lactamase production, 7.6% of the 53 encapsulated strains in Manitoba produced β-lactamase and were resistant to ampicillin.[26] Similar percentages of non-typeable isolates showing resistance to ampicillin due to β-lactamase production were found in both British Columbia and Manitoba. Resistance to cefaclor and clarithromycin was rare in both British Columbia and Manitoba, but resistance to trimethoprim–sulfamethoxazole was common in both provinces. The most dramatic difference observed between these two provinces was the much higher rate of BLNAR genotypes in the invasive non-typeable *H. influenzae* in British Columbia (31%) compared to Manitoba (13%). Also, there were more β-lactamase producing isolates with significant *ftsI* gene mutations in British Columbia (4.6%) than in Manitoba (1.4%). A unique non-typeable isolate was found in British Columbia that showed resistance to multiple antibiotics, including chloramphenicol, tetracycline, ampicillin, and trimethoprim–sulfamethoxazole, and had significant mutations in the *ftsI* gene. The higher number of genotypic BLNAR and BLPACR isolates in British Columbia may reflect an increase in prevalence of such isolates over time rather than a geographical difference, as the Manitoba data were based on isolates collected between 1990 and 2006.[34] Regardless of the cause of this difference, the finding suggests that continuous monitoring by laboratory surveillance is important to track the development of resistance to β-lactam antibiotics in *H. influenzae*.

Besides documenting an evolving *H. influenzae* population, the demographics of patients with invasive *H. influenzae* disease has also changed enormously in the last two decades since the introduction of Hib vaccines. Instead of being mainly a pediatric disease, different capsular types and non-encapsulated *H. influenzae* appear to be causing disease in patients of different age groups. In 1989, before Hib conjugate vaccines were introduced in Canada, 24 cases of invasive Hib were reported in the province of British Columbia. Twenty-one (88%) of these 24 cases were in children aged 14 years or below and 17 cases (71%) were in children less than 5 years old. There were only three cases in adults over the age of 25 years (Public Health Agency of Canada, Notifiable Diseases On-line: http://dsol-smed.hc-sc.gc.ca/dsol-smed/ndis/index-eng.php). In 2008 and 2009, 75% of the 12 Hib cases were in adults aged 21 years and above. It appears that serotype a *H. influenzae* has now replaced Hib as the most common cause of invasive disease in Canadian children aged 2 years or under, with 80% of the cases in British Columbia (this study) and 72% of the Manitoba cases[12] in this age group. More common than serotype a are non-typeable or non-encapsulated strains as the cause of invasive disease; and in British Columbia, these strains appear to affect adults more than children (83% of invasive non-typeable cases were in those over the age of 18 years). It is not known what causes patients of a certain age group to be susceptible to certain strains of invasive *H. influenzae*. It could be the immunological status of the host or perhaps encapsulation and/or other genetic makeup of the *H. influenzae* strains. This is an intriguing question that is common to many if not all encapsulated invasive bacterial disease agents and the infectious process; answers to such a complex question would require more carefully planned studies of both the pathogen as well as the host.

This was a retrospective laboratory surveillance study, therefore there are some limitations to the data observed. While Hib has been a reportable disease in Canada since 1979, non-Hib cases have only been included in the revised national notifiable disease list since 2007.[35] Therefore, the number of non-Hib cases may potentially be an underestimate, as not all laboratories may be familiar with this change and some may not have submitted the non-Hib isolates for further typing. Also we assumed all patients were residents of British Columbia and did not consider the possibility that some patients might have been visitors from other provinces. We did not look at the rates of disease caused by the different serotypes or non-typeable strains, and therefore, it was not possible to determine if there was an overall increase in invasive *H. influenzae* disease or only an increase in the proportion of non-Hib strains identified from invasive disease cases. Finally, we have no historical data on non-Hib disease and it is not possible to determine if the number of invasive non-Hib disease cases observed now has changed compared to the number in the past. However, from a casual comparison of our current data from this study (with 45–53 cases per year) with the data obtained in 1989 before the Hib conjugate vaccine era (24 cases of Hib), it may be reasonable to conclude that Hib as a cause of invasive diseases in children in the past has now been replaced by non-Hib strains causing invasive disease in adults and serotype a strains causing disease in young children.

Conflict of interest

None of the authors have any conflicts of interest to declare.

Acknowledgements

The authors would like to thank co-op student Ashley Muckle for assistance with antibiotic susceptibility testing. We also thank the staff at the National Microbiology Laboratory's DNA Core Facility for the DNA sequencing work. R. Tsang has received funding from Health Canada's Biotechnology-Genomics Research and Development Fund for studies on vaccine preventable bacterial diseases. Ethics approval for this study was not required.

This study made use of the *Haemophilus influenzae* MLST website (http://www.haemophilus.mlst.net), developed and maintained by David Aanensen at the Imperial College, London, UK, and funded by the Wellcome Trust. The site is currently curated by Daniel Godoy.

References

1. Pittman M. Variation and type specificity in the bacterial species *Haemophilus influenzae*. *J Exp Med* 1931;**53**:471–92.
2. Falla TJ, Dobson SR, Crook DW, Kraak WA, Nichols WW, Anderson EC, et al. Population-based study of non-typeable *Haemophilus influenzae* invasive disease in children and neonates. *Lancet* 1993;**341**:851–4.
3. Mason Jr EO, Kaplan SL, Lamberth LB, Hinds DB, Kvernland SJ, Loiselle EM, et al. Serotype and ampicillin susceptibility of *Haemophilus influenzae* causing systemic infections in children: 3 years of experience. *J Clin Microbiol* 1982;**15**:543–6.
4. Wenger JD, Ward JI. Haemophilus influenzae vaccine. In: Plotkin SA, Orenstein W, editors. *Vaccines*. 4th ed., Philadelphia, PA: WB Saunders Co; 2004 p. 229–68.
5. Bath S, Bisgard K, Murphy T, Shutt K, Rosenstein N, Ohuabunwo C. Progress toward elimination of *Haemophilus influenzae* type b invasive disease among infants and children—United States, 1998-2000. *MMWR Morb Mortal Wkly Rep* 2002;**51**:234–7.
6. Scheifele D, Halperin S, Vaudry W, Jadavji T, Tan B, Law B, et al. Historic low Haemophilus influenzae type b case tally—Canada 2000. *Can Commun Dis Rep* 2001;**27**:149–50.

7. Farley MM, Stephens DS, Brachman PS, Harvey C, Smith JD, Wenger JD. Invasive *Haemophilus influenzae* disease in adults. CDC Meningitis Surveillance Group. *Ann Intern Med* 1992;**116**:806–12.
8. Lipsitch M. Vaccination against colonizing bacteria with multiple serotypes. *Proc Natl Acad Sci U S A* 1997;**94**:6571–6.
9. Tsang RS. Capsule switching and capsule replacement in vaccine preventable bacterial diseases. *Lancet Infect Dis* 2007;**7**:569–70.
10. Tsang RS. The changing epidemiology of invasive *Haemophilus influenzae* disease. *Lancet Infect Dis* 2008;**8**:737.
11. Dworkin MS, Park L, Borchardt SM. The changing epidemiology of invasive *Haemophilus influenzae* disease, especially in persons ≥65 years old. *Clin Infect Dis* 2007;**44**:810–6.
12. Tsang RS, Sill ML, Skinner SJ, Law DK, Zhou J, Wylie J. Characterization of invasive *Haemophilus influenzae* disease in Manitoba, Canada, 2000–2006: invasive disease due to non-type b strains. *Clin Infect Dis* 2007;**44**:1611–4.
13. Brown VM, Madden S, Kelly L, Jamieson FB, Tsang RS, Ulanova M. Invasive *Haemophilus influenzae* disease caused by non-type b strains in Northwestern Ontario, Canada, 2002–2008. *Clin Infect Dis* 2009;**49**:1240–3.
14. Adam HJ, Richardson SE, Jamieson FB, Rawte P, Low DE, Fisman DN. Changing epidemiology of invasive *Haemophilus influenzae* in Ontario, Canada: evidence for herd effects and strain replacement due to Hib vaccination. *Vaccine* 2010;**28**:4073–8.
15. Kilian M. Haemophilus. In: Murray PR, Jo Baron E, Jorgensen JH, Pfaller MA, Yolken RH, editors. *Manual of clinical microbiology*. 8th ed., Washington, DC: ASM Press; 2003. p. 623–35.
16. Lau SK, Woo PC, Mok MY, Teng JL, Tam VK, Chan KK, et al. Characterization of *Haemophilus segnis*, an important cause of bacteremia, by 16S rRNA gene sequencing. *J Clin Microbiol* 2004;**42**:877–80.
17. Kilian M. A taxonomic study of the genus *Haemophilus*, with the proposal of a new species. *J Gen Microbiol* 1976;**93**:9–62.
18. Falla TJ, Crook DW, Brophy LN, Maskell D, Kroll JS, Moxon ER. PCR for capsular typing of *Haemophilus influenzae*. *J Clin Microbiol* 1994;**32**:2382–6.
19. Meats E, Feil EJ, Stringer S, Cody AJ, Goldstein R, Kroll JS, et al. Characterization of encapsulated and non-encapsulated *Haemophilus influenzae* and determination of phylogenetic relationship by multilocus sequence typing. *J Clin Microbiol* 2003;**41**:1623–36.
20. Clinical and Laboratory Standards Institute. Performance standards for antimicrobial susceptibility testing. Eighteenth informational supplement. M100-S18. Wayne, PA: CLSI; 2008.
21. Karpanoja P, Nissinen A, Huovinen P, Sarkkinen H. Disc diffusion susceptibility testing of *Haemophilus influenzae* by NCCLS methodology using low-strength ampicillin and co-amoxiclav disc. *J Antimicrob Chemother* 2004;**53**:660–3.
22. Ubukata K, Shibasaki Y, Yamamoto K, Chiba N, Hasegawa K, Takeuchi Y, et al. Association of amino acid substitutions in penicillin-binding protein 3 with β-lactam resistance in β-lactamase-negative ampicillin-resistant *Haemophilus influenzae*. *Antimicrob Agents Chemother* 2001;**45**:1693–9.
23. Fleischmann RD, Adams MD, White O, Clayton RA, Kirkness EF, Kerlavage AR, et al. Whole-genome random sequencing and assembly of *Haemophilus influenzae* Rd. *Science* 1995;**269**:496–512.
24. Dabernat H, Delmas C, Seguy M, Pelissier R, Faucon G, Bennamani S, et al. Diversity of β-lactam resistance-conferring amino acid substitutions in penicillin-binding protein 3 of *Haemophilus influenzae*. *Antimicrob Agents Chemother* 2002;**46**:2208–18.
25. Bajanca P, Canica M, the Multicenter Study Group. Emergence of nonencapsulated and encapsulated non-b type invasive Haemophilus influenzae isolates in Portugal (1989–2001). *J Clin Microbiol* 2004;**42**:807–10.
26. Sill ML, Law DK, Zhou J, Skinner S, Wylie J, Tsang RS. Population genetics and antibiotic susceptibility of invasive Haemophilus influenzae in Manitoba, Canada, from 2000 to 2006. *FEMS Immunol Med Microbiol* 2007;**51**: 270–6.
27. Schouls LM, van der Ende A, van de Pol I, Schot C, Spanjaard L, Vauterin P, et al. Increase in genetic diversity of *Haemophilus influenzae* serotype b (Hib) strains after introduction of Hib vaccination in the Netherlands. *J Clin Microbiol* 2005;**43**:2741–9.
28. Bruun B, Gahrn-Hansen B, Westh H, Kilian M. Clonal relationship of recent invasive *Haemophilus influenzae* serotype f isolates from Denmark and the United States. *J Med Microbiol* 2004;**53**:1161–5.
29. Bruce MG, Deeks SL, Zulz T, Navarro C, Palacios C, Case C, et al. Epidemiology of *Haemophilus influenzae* serotype a, North American Arctic, 2000–2005. *Emerg Infect Dis* 2008;**14**:48–55.
30. Millar EV, O'Brien KL, Watt JP, Lingappa J, Pallipamu R, Rosenstein N, et al. Epidemiology of invasive *Haemophilus influenzae* type a disease among Navajo and White Mountain Apache children, 1988–2003. *Clin Infect Dis* 2005;**40**: 823–30.
31. McConnell A, Tan B, Scheifele D, Halperin S, Vaudry W, Law B, et al. Invasive infections caused by *Haemophilus influenzae* serotypes in twelve Canadian IMPACT centers, 1996–2001. *Pediatr Infect Dis J* 2007;**26**:1025–31.
32. Kroll JS, Moxon ER, Loynds BM. Natural genetic transfer of a putative virulence enhancing mutation to *Haemophilus influenzae* type a. *J Infect Dis* 1994;**169**:76–9.
33. Adderson EE, Byington CL, Spencer L, Kimball A, Hindiyeh M, Carroll K, et al. Invasive serotype a Haemophilus influenzae infections with a virulence genotype resembling Haemophilus influenzae type b: emerging pathogen in the vaccine era? *Pediatrics* 2001;**108**:e18–23.
34. Sill ML, Tsang RS. Antibiotic susceptibility of invasive *Haemophilus influenzae* strains in Canada. *Antimicrob Agents Chemother* 2008;**52**:1551–2.
35. Public Health Agency of Canada. Case definitions for communicable diseases under national surveillance. Can Commun Dis Rep 2009; 35 (Suppl 2).